FLATNESS OF A COLIMIT MODULE IS REALISED BY ONE STAGE OF THE SYSTEM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The local criterion for flatness over a local ring. Invoked at the end of the counted proof (lines 32974-32976). Turns the vanishing of a Tor group over the residue field into flatness of the stage.
- Commutation of tensor products with colimits. Used in the counted proof at lines 32959-32963. Moves the vanishing of the finitely many generators to a later stage of the system.

Source reference. Original lemma 00MO: sources/stacks-algebra/pinned-original-source.tex, source lines 24138–24138. Navigation pointer for \ref{lemma-another-variant-local-criterion-flatness} used inside the counted span (line 32977); the argument that consumes it is retained in the span.

Source reference. Original lemma 00QX: sources/stacks-algebra/pinned-original-source.tex, source lines 32644–32644. Navigation pointer for \ref{lemma-limit-module-essentially-finite-presentation} used inside the counted span (line 32942); the argument that consumes it is retained in the span.

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Source reference. Original lemma 00M6: sources/stacks-algebra/pinned-original-source.tex, source lines 18907–18907. Navigation pointer for \ref{remark-Tor-ring-mod-ideal} used inside the counted span (line 32956); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$, M, Λ, $R_\lambda \to S_\lambda$, M_λ be as in Lemma 00QX. Assume that M is flat over R. Then for some $\lambda \in \Lambda$ the module M_λ is flat over R_λ.*

Proof. Pick some $\lambda \in \Lambda$ and consider

$$\mathrm{Tor}_1^{R_\lambda}(M_\lambda, R_\lambda/\mathfrak{m}_\lambda) = \mathrm{Ker}(\mathfrak{m}_\lambda \otimes_{R_\lambda} M_\lambda \to M_\lambda).$$

See Remark 00M6. The right hand side shows that this is a finitely generated S_λ-module (because S_λ is Noetherian and the modules in question are finite). Let $\xi_1, \ldots, \xi_n$ be generators. Because M is flat over R we have that $0 = \mathrm{Ker}(\mathfrak{m}_\lambda R \otimes_R M \to M)$. Since $\otimes$ commutes with colimits we see there exists a $\lambda' \geq \lambda$ such that each ξ_i maps to zero in $\mathfrak{m}_\lambda R_{\lambda'} \otimes_{R_{\lambda'}} M_{\lambda'}$. Hence we see that

$$\mathrm{Tor}_1^{R_\lambda}(M_\lambda, R_\lambda/\mathfrak{m}_\lambda) \longrightarrow \mathrm{Tor}_1^{R_{\lambda'}}(M_{\lambda'}, R_{\lambda'}/\mathfrak{m}_\lambda R_{\lambda'})$$

is zero. Note that $M_\lambda \otimes_{R_\lambda} R_\lambda/\mathfrak{m}_\lambda$ is flat over $R_\lambda/\mathfrak{m}_\lambda$ because this last ring is a field. Hence we may apply Lemma 00MO to get that $M_{\lambda'}$ is flat over $R_{\lambda'}$. $\square$